

TECHHUB ELECTRONICS

TECHNOLOGY ASSESSMENT REPORT

Technology choices for a connected, omnichannel retail operating model

BCS CAPSTONE PROJECT

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Prepared for the Executive Steering Committee

Planning basis: 65 stores | ~\$280m revenue | 850 employees | \$18.6m three-year programme

Decision purpose Evaluate CRM, ERP, forecasting and inventory AI, e-commerce, POS, BI and inventory/WMS options; define the integrated target stack; and establish the procurement and implementation controls needed to deliver the approved 36-month roadmap.

EXECUTIVE BRIEF

Contents and decision at a glance

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Executive recommendation

Adopt a Microsoft-centred composable retail stack:

Dynamics 365 Commerce and Store Commerce;

Dynamics 365 Supply Chain Management for WMS and inventory;

Distributed Order Management plus Inventory Visibility for OMS/ATP;

Customer Service and Customer Insights for CRM/consent;

Power BI/Fabric for analytics; and

Azure integration,

identity and monitoring services.

Use Dynamics 365 Demand Planning first, with RELEX retained as a specialist contingency if the controlled forecast pilot does not beat the validated baseline.

20% digital revenue M36	98.5% warehouse accuracy	99.5% stock messages	AU\$23.4m five-year TCO
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1. TECHNOLOGY LANDSCAPE ANALYSIS

1.1 Current technology assessment

The current environment has not been fully inventoried. The assessment therefore separates established programme facts from discovery assumptions. No vendor contract should be awarded until product versions, interfaces, volumes, support status, data ownership and failure history are verified in Months 1-3.

Domain	Current known / planning assumption	Primary constraint	Urgency
Digital commerce	Online sales are 8%; current experience is not providing dependable product, stock and delivery information.	Conversion, accessibility, speed and order-status gaps.	Critical
Order + inventory	Store and warehouse availability is fragmented; reliable ATP and reservations are not yet enterprise-wide.	Overselling, manual checks and inconsistent promises.	Critical
Warehouse	Processes need scanning, directed work, bin and serial control; device and Wi-Fi readiness must be confirmed.	Accuracy, cycle time and exception visibility.	Critical
POS / stores	65 stores require common product, price, order, customer and stock information; current vendor/version is TBC.	Store service must not be disrupted by fulfilment work.	High
CRM / loyalty	Customer, consent, purchase and service history are not yet unified.	Privacy, duplicate identities and hand-offs.	High
ERP / finance	Underlying finance, procurement and ERP applications are not yet evidenced.	Replacing an unknown core would create avoidable cost and risk.	Assess
Data / integration / BI	Point-to-point reliability and data definitions are insufficient; baselines need validation.	No trusted end-to-end monitoring or benefits evidence.	Critical
Security / resilience	Control maturity, PCI scope, recovery results and privileged-access design require evidence.	Launch gates cannot rely on assumed compliance.	Critical

1.2 Assessment conclusion

The binding technology problem is orchestration and trusted data, not simply replacing the website.

A coherent suite reduces interfaces, but specialist alternatives remain viable where warehouse or planning complexity justifies them.

Architecture, data ownership, operating support and adoption must be contracted as deliverables, not treated as later activities.

Discovery output A verified application register, interface catalogue, data profile, volume model, licensing baseline, cyber/privacy gap assessment and signed process maps are mandatory inputs to the RFP shortlist.

2. TECHNOLOGY REQUIREMENTS

2.1 Functional requirements

The selection must support the full customer and fulfilment process

1 Product + price Approved master data	2 Discover Web / store search	3 Promise ATP + reservation	4 Order + pay Checkout / POS
5 Route OMS rules	6 Fulfil WMS / store	7 Deliver Ship / collect	8 Serve + return Track / case / refund

MoSCoW priorities

Priority	Required capability	Acceptance evidence
Must	Single product/price definitions; store + DC inventory; ATP dates; reservations; safety stock; controlled split orders.	Live scenario test across representative store and DC stock states.
Must	Accessible responsive commerce; secure payment; accounts; tracking; delivery/collection; cancellations and returns.	WCAG 2.2 AA audit, load test, payment assurance and journey tests.
Must	OMS routing by stock, distance, capacity, cutoff, restrictions, time and cost; observable exceptions and replay.	Golden-order scenarios, failure injection and audit trace.
Must	WMS receiving, serial scan, directed putaway, bins, replenishment, pick/pack/dispatch, counts, holds and returns.	Warehouse pilot at required accuracy and productivity thresholds.
Must	POS order lookup, digital receipts, cross-location ordering, collection proof, returns and controlled stock adjustments.	Store pilot with customer, accuracy and labour thresholds.
Should	Unified customer profile, consent, loyalty, communications, purchases and service cases.	Identity matching, consent reconciliation and role-access test.
Should	Approved KPI semantic model, operational dashboards, forecast pilots and planner-approved replenishment suggestions.	Finance reconciliation and model back-test against baseline.
Could	Advanced personalisation, marketplace channels and separate mobile app after core stability.	New business case after Month 30 evidence.

2.2 Non-functional requirements

Website availability >=99.95%; normal stock updates within five minutes; >=99.5% stock-message processing; elastic peak performance; Australian data/privacy assessment; encryption in transit and at rest; least privilege and MFA; immutable audit evidence; automated retry/replay and reconciliation; recovery objectives proven by test; accessible role-based interfaces; versioned APIs; and exportable data with documented exit assistance.

INTEGRATION ARCHITECTURE

2.3 Target logical architecture

Systems of engagement remain decoupled from orchestration, execution and record

Experience	Digital commerce	Store Commerce / POS	Service + CRM
Orchestration	OMS + ATP + reservations	Payments / fraud	Notifications / returns
Execution	WMS + devices	Store fulfilment	Carriers / collection
Core records	ERP / finance	Product + price master	Customer / consent master
Platform	API management + events	Data platform + BI	Identity + monitoring

Authoritative data ownership

Data object	System of record	Key consumers	Control
Product + price	ERP/product master; approved channel publication	Web, POS, CRM, OMS, WMS	Named owner, effective dates, approval, reconciliation
Customer + consent	CRM / Customer Insights	Web, POS, service, marketing	Identity matching, purpose, channel, timestamp, opt-out
Physical stock	WMS / store inventory ledger	Inventory Visibility, OMS, POS, BI	Scan/transaction source, reason code, sequence, count
Available-to-promise	OMS / availability service	Web, POS, service	Reservations, safety stock, capacity, cutoff and expiry
Order	OMS	Web, POS, WMS, stores, CRM, finance	Idempotency, status model, event history, exception owner
Financial posting	Finance ERP	OMS, BI, audit	Balanced posting, settlement and period controls

Integration pattern

Synchronous APIs only for immediate customer promises and commands; asynchronous events for state propagation and scale.

Canonical schemas, correlation IDs, idempotency keys, dead-letter queues, safe replay and business reconciliation for every critical flow.

One observability view spans commerce, payment, OMS, WMS/store, carrier, CRM and finance; alerts name the business impact and owner.

Dependency sequence Integration platform and master-data rules start before channel builds. OMS/ATP must be proven before broad click-and-collect. WMS/store execution must be stable before scale. BI and AI consume governed events and reconciled history; they do not repair weak source processes.

3. VENDOR EVALUATION

3.1 CRM systems

Customer profile, consent, service, loyalty and order context

Weights: functional fit 30%; integration/architecture 20%; implementation complexity 15%; five-year TCO 15%; security/scalability 10%; ecosystem/support 10%. Scores are preliminary (1-5) and must be confirmed through scripted demonstrations, reference calls, security due diligence and commercial bids.

Option	Relevant capability and TechHub fit	Five-year ROM*	Complexity	Score /5
Microsoft Dynamics 365 Customer Service + Customer Insights	Strong fit with recommended Commerce, Dataverse, identity and Power BI stack; supports service cases, unified profiles, consent-aware journeys and role-based access.	\$1.8-2.6m	Medium	4.4
Salesforce Service Cloud + Data Cloud	Mature service and customer-data ecosystem with broad partners; strong option if customer-service sophistication outweighs the added integration layer.	\$2.4-3.6m	Med-high	4.3
HubSpot Enterprise	Fast adoption and lower administration burden; weaker fit for complex retail identity, service, loyalty and fine-grained enterprise integration at TechHub scale.	\$1.4-2.2m	Low-med	3.5

Recommendation and proof points

Select Dynamics 365 Customer Service and Customer Insights, subject to proof that consent, customer matching, loyalty integration, order visibility and service workflows work across website, POS and contact-centre roles. Salesforce is the strongest functional alternative; HubSpot is a cost-led option but would need careful boundary design.

RFP test	Pass condition
Customer 360	Agent sees authorised preferences, consent, purchases, orders, returns, loyalty and cases without switching systems.
Identity + consent	Duplicate test records merge/hold correctly; opt-out reaches all channels within the agreed time and remains auditable.
Service recovery	Agent can explain status, cancel within rules, initiate return and record outcome with least privilege.
Exit	TechHub receives documented export of profiles, cases, consent evidence and configuration in usable formats.

ROM note Category ranges are TechHub planning estimates in nominal AU\$, excluding GST. They include implementation and run costs but overlap with shared platform/integration costs; do not add them to derive programme TCO.

VENDOR EVALUATION

3.2 ERP solutions

Finance, procurement, product and supply-chain system-of-record options

Option	Relevant capability and TechHub fit	Five-year ROM*	Complexity	Score /5
Microsoft Dynamics 365 Finance + SCM	Best architectural alignment with the proposed retail stack and Power Platform. Strong supply-chain fit; finance replacement still requires a separate fit-gap and business case.	\$4.8-7.0m	High	4.4
Oracle Fusion Cloud ERP + SCM	Broad global finance, procurement and supply-chain scope; capable but introduces another platform and a higher integration/change burden.	\$6.0-9.0m	High	4.1
SAP S/4HANA Cloud	Deep enterprise process control and supply-chain capability; likely disproportionate complexity and cost for TechHub unless the current landscape is already SAP-centred.	\$6.5-10.0m	Very high	4.0

Recommendation: target alignment, not automatic replacement

Set Dynamics 365 Finance and SCM as the preferred target architecture, but procure only the SCM/WMS, inventory and retail capabilities needed by the approved roadmap in the first wave. Retain and integrate the present finance core if it remains supported, secure and capable of timely postings, tax, settlement and management reporting.

Month 1-3 fit-gap question	Decision rule
Supportability and risk	Replace only if end-of-support, material control weakness or unacceptable operational failure cannot be remediated.
Functional fit	Test finance, procurement, product, vendor, landed-cost, serial/warranty and returns-accounting requirements.
Integration economics	Compare five-year coexistence cost with replacement cost, including interfaces, duplicated data, testing and internal support.
Cutover feasibility	Require reconciled opening balances, open items, stock valuation, tax and settlement rehearsal with rollback.

Funding boundary The approved \$18.6m three-year programme does not include a broad finance ERP transformation. If replacement is required, the steering committee must approve added scope, benefits, contingency and schedule rather than silently consuming customer and fulfilment funding.

VENDOR EVALUATION

3.3 E-commerce platforms

Accessible discovery, checkout and post-purchase journeys

Option	Relevant capability and TechHub fit	Five-year ROM*	Complexity	Score /5
Microsoft Dynamics 365 Commerce	Native alignment with Store Commerce, product/price, loyalty and Microsoft operations stack; supports a unified retail model but requires disciplined implementation and front-end performance engineering.	\$3.5-4.8m	Med-high	4.4
Adobe Commerce	Strong merchandising, extensibility and complex catalogue capability; attractive for differentiated digital experience but higher engineering and integration overhead.	\$4.2-6.2m	High	4.3
Shopify Plus	Fast SaaS delivery, strong ecosystem and approachable operations; enterprise OMS, complex pricing and store-process integration may require more third-party components.	\$2.4-3.8m	Medium	4.0

Recommendation and scope

Select Dynamics 365 Commerce as the preferred channel platform because the main business outcome depends on consistent price, ATP, order and loyalty information across web and stores. Use a decoupled front end only if performance testing shows a material advantage and TechHub can operate the added release and observability burden.

Release 1 included	Deferred until stable
Responsive search, compare, product information and availability	Separate native mobile application
Guest/account checkout, secure card payment and fraud integration	Marketplace selling
Delivery/collection promise, tracking, cancellation and returns guidance	Advanced AI shopping assistant
Accessibility, speed, analytics, content migration and SEO controls	Complex personalisation without consent and measured value

Acceptance thresholds

Key journeys pass WCAG 2.2 AA; monthly availability reaches $\geq 99.95\%$; peak load tests meet agreed response thresholds.

Displayed promise equals reserved capacity and inventory; payment retries never create duplicate orders or charges.

Digital revenue reaches 20%, conversion improves 60% from the validated baseline and CSAT rises 12 points by Month 36.

VENDOR EVALUATION

3.4 Point-of-sale modernisation

Unified retail transactions while protecting in-store service

Option	Relevant capability and TechHub fit	Five-year ROM*	Complexity	Score /5
Microsoft Dynamics 365 Store Commerce	Best fit with the proposed Commerce, pricing, loyalty, customer and order stack; supports selective modernisation rather than simultaneous replacement of every POS component.	\$2.2-3.2m	Med-high	4.5
Oracle Retail Xstore	Mature enterprise retail POS with omnichannel and inventory functions; strong alternative, but introduces an Oracle retail layer and additional integration/accountability boundaries.	\$3.0-4.5m	High	4.2
Shopify POS	Simple administration and close alignment with Shopify commerce; less suitable where complex electronics returns, serial/warranty controls and enterprise store orchestration dominate.	\$1.8-2.8m	Medium	3.8

Recommendation and rollout guardrails

Use Store Commerce where POS change is required, but first test whether the current POS can be securely integrated for core omnichannel functions. A selective replacement path reduces disruption and preserves investment while still giving stores common product, price, customer, stock and order information.

Store capability	Pilot control
Order search, digital receipt and cross-location ordering	No material increase in checkout time or training-related queue length.
Collection task queue, handheld scan, staging and proof	>=95% ready on time; secure staging and customer verification pass.
Cross-channel return and refund	Original tender, serial/warranty, fraud and financial posting rules reconcile.
Stock adjustment	Reason-coded, role-controlled and reflected in ATP normally within five minutes.
Workload protection	Capacity caps prevent fulfilment tasks from degrading normal customer service.

Pilot selection Choose representative high/medium-volume stores, different formats and locations, adequate network/device readiness and credible local leadership. Scale only after accuracy, customer, labour and support thresholds pass for a sustained period.

VENDOR EVALUATION

3.5 Inventory management, WMS and OMS/ATP

Accurate physical execution and profitable network-wide promises

Option	Relevant capability and TechHub fit	Five-year ROM*	Complexity	Score /5
Dynamics 365 SCM WMS + Inventory Visibility	Directed warehouse work, bins, serial capture, inventory visibility and close fit with the recommended stack. Suitable if discovery confirms moderate DC automation and complexity.	\$4.5-6.5m	High	4.3
Manhattan Active Warehouse Management	Cloud-native specialist depth, real-time operations and advanced fulfilment; strongest functional option for high automation or complex distribution, with higher cost and integration burden.	\$5.5-8.0m	Very high	4.4
Infor WMS	Strong tier-one WMS capability, labour/voice/analytics options and broad supply-chain fit; adds a distinct platform and specialist operating model.	\$4.8-7.0m	High	4.2

OMS/ATP overlay - three viable options

Option	Fit	Score /5
D365 Commerce DOM + Inventory Visibility	Preferred: lowest integration burden with recommended commerce, Store Commerce and SCM; validate routing scale, reservations and exceptions.	4.4
Manhattan Active Order Management	Deep precise promising, orchestration and store fulfilment; preferred contingency if rules and volume exceed suite capability.	4.3
Fluent Order Management	Composable real-time inventory and workflow approach; attractive with heterogeneous channels, but introduces another vendor and support boundary.	4.1

Recommendation and conditional gate

Select Dynamics 365 SCM WMS plus Commerce DOM and Inventory Visibility, conditional on a warehouse and golden-order proof. Reopen Manhattan for WMS/OMS if discovery finds material automation, wave/labour complexity, very high transaction peaks or promise logic that the Microsoft design cannot meet without extensive customisation.

Pilot thresholds: warehouse inventory $\geq 98.5\%$; order accuracy $\geq 99.2\%$; cycle time -30%; pick productivity +20%.

Network thresholds: all active locations in ATP; $\geq 99.5\%$ stock updates processed; normal updates ≤ 5 minutes; pickup ready $\geq 95\%$.

VENDOR EVALUATION

3.6 Data analytics and business intelligence

One governed measurement layer for operations, customers and benefits

Option	Relevant capability and TechHub fit	Five-year ROM*	Complexity	Score /5
Microsoft Power BI + Fabric	Strongest end-to-end fit with Azure/Dynamics, broad analyst adoption and competitive public user pricing; requires semantic-model and capacity governance.	\$2.0-3.0m	Medium	4.6
Tableau Cloud	Excellent visual exploration and mature analytics community; added data-platform integration and separate administration reduce stack coherence.	\$2.4-3.6m	Medium	4.1
Qlik Cloud Analytics	Strong associative analysis and governed self-service; capable alternative, but less aligned with the proposed application and identity platform.	\$2.4-3.7m	Medium	3.9

Recommendation and operating model

Adopt Power BI and Fabric, with one Finance-approved semantic model for programme KPIs. Separate certified management measures from exploratory analysis. Capacity, refresh, export, row-level security and data-retention rules must be designed before broad self-service rollout.

Phase	Analytics product	Evidence
1 M1-9	Baseline and data-quality scorecards	Definitions, owners, source-to-report lineage and Finance reconciliation.
2 M9-24	Order, inventory, warehouse, store, customer and benefits dashboards	Operational use, alert ownership, adoption and monthly control review.
3 M24-36	Forecast monitoring, replenishment suggestions and measured personalisation	Back-tests, control group, drift, override reasons and realised value.

+15% inventory turnover	-40% stockouts	+12 pts customer satisfaction	>=90% role proficiency
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Measurement rule No benefit is claimed from dashboard activity alone. Each measure requires an approved baseline, source reconciliation, named owner, calculation logic, frequency and link to a realised financial or service outcome.

VENDOR EVALUATION

3.7 AI for demand forecasting and inventory optimisation

Controlled decision support after data and process stability

Option	Relevant capability and TechHub fit	Five-year ROM*	Complexity	Score /5
Dynamics 365 Demand Planning	Integrated collaborative planning, best-fit/custom forecasting and Copilot-supported analysis; lowest-friction starting point for the recommended stack.	\$1.8-2.8m	Med-high	4.1
RELEX Forecasting + Replenishment	Retail-specialist depth across forecasting, replenishment and inventory optimisation; strongest functional score, but higher data/integration and operating-model cost.	\$3.0-5.0m	High	4.4
Blue Yonder Demand Planning + Inventory Optimisation	Deep enterprise supply-chain planning and ML capability; powerful but likely more platform and change complexity than the first TechHub use cases require.	\$3.5-5.5m	Very high	4.2

Recommendation: staged Microsoft pilot with a specialist exit gate

Begin with Dynamics 365 Demand Planning in Months 24-30 because it minimises integration and supports planner collaboration. Do not automate material replenishment decisions during the pilot. If the solution fails the agreed accuracy, bias, service and usability thresholds, run a time-boxed RELEX proof using the same frozen dataset and scoring method.

Control	Required evidence
Baseline	Compare against the current approved method by category/store horizon; use rolling back-tests, not one favourable period.
Decision rights	Planners approve important orders; overrides capture reason, expected impact and outcome.
Model risk	Measure forecast error, bias, stockout/service impact, stability by sparse/new items, drift and failure handling.
Privacy + fairness	Personalisation uses authorised data and opt-out; avoid sensitive inference and document purpose limitations.
Value	Proceed only where margin, availability, working capital and labour gains exceed license, data and operating cost.

No premature AI Advanced shopping assistants and autonomous inventory decisions remain outside Release 1. Trustworthy product, stock, order and customer data—and reliable human processes—are prerequisites, not parallel assumptions.

5. RECOMMENDATIONS

5.1 Recommended technology stack

A coherent default with explicit specialist contingency gates

Layer	Preferred component	Why it is recommended	Condition / alternative
Channels	D365 Commerce + Store Commerce	Common price, loyalty, customer and order context across web and stores.	Adobe/Shopify if digital differentiation or SaaS speed materially outweighs integration.
Order promise	Commerce DOM + Inventory Visibility	Matches routing, reservation and ATP needs with the lowest suite boundary.	Manhattan or Fluent if golden-order/scale tests fail.
Warehouse + inventory	D365 SCM WMS	Scanning, serials, directed work, bins and inventory in the preferred operations platform.	Manhattan if automation/complexity requires specialist depth.
CRM	D365 Customer Service + Customer Insights	Unified customer, service, consent and loyalty integration with the retail stack.	Salesforce if service sophistication produces demonstrably higher value.
ERP	D365 Finance/SCM target architecture	Coherent long-term platform and data ownership.	Retain current finance core unless fit-gap justifies funded replacement.
Analytics	Power BI + Fabric	Governed semantic model and close identity/application alignment.	Tableau/qlik only if capability proof offsets platform fragmentation.
Planning AI	D365 Demand Planning pilot	Lowest-friction controlled start with human approval.	RELEX proof if Microsoft pilot fails frozen test.
Integration + security	Azure API Management, Service Bus/Event Grid, Logic Apps, Monitor, Entra ID, Purview	API/event backbone, identity, observability and governance across the stack.	Use equivalent managed services if current enterprise standards mandate them.

Detailed justification

Business fit: directly supports the approved process from reliable product/price through ATP, OMS routing, WMS/store fulfilment, service and returns.

Delivery fit: suite coherence reduces custom interfaces and shared accountability while preserving specialist alternatives at explicit gates.

Financial fit: the stack can be staged inside the \$18.6m three-year envelope if finance ERP replacement is excluded and scope is governed.

Operating fit: Microsoft skills are broadly available; one identity, monitoring and data ecosystem lowers long-term support fragmentation.

Risk fit: pilots, wave rollouts, rollback plans and output-based contracts expose limitations before enterprise scale.

Procurement posture Negotiate architecture and outcomes together: environments, interfaces, data migration, test automation, performance, accessibility, security evidence, observability, knowledge transfer, exit support and cost escalation protections must be part of the contracted solution.

COMMERCIAL ASSESSMENT

3.8 Total cost of ownership - five-year projection

Nominal AU\$ millions, excluding GST; steering baseline, not vendor quotation

Workstream	Y1-3 build + rollout	Y4	Y5	5-year
Digital commerce	3.00	0.35	0.35	3.70
OMS / ATP	2.50	0.25	0.25	3.00
WMS + devices/infrastructure	4.00	0.35	0.35	4.70
Integration, data + monitoring	2.40	0.45	0.45	3.30
POS + store enablement	2.00	0.25	0.25	2.50
Security, resilience + change	1.60	0.20	0.20	2.00
CRM + loyalty	1.50	0.25	0.25	2.00
BI + forecasting/personalisation	1.60	0.30	0.30	2.20
Total	18.60	2.40	2.40	23.40

Annual cash-flow planning

Year 1	Year 2	Year 3	Year 4	Year 5	Total
6.20	8.10	4.30	2.40	2.40	23.40

Included / excluded and sensitivity

Included	Excluded / separate approval
Subscription/license allowances; implementation; integration; data migration; environments; devices; testing; assurance; training; pilot support; initial optimisation; ongoing run allowance.	GST; acquisition-related changes; major DC automation/building works; broad network refresh outside initiative needs; full finance ERP replacement; marketplace/mobile app; advanced AI scale.

Scenario	Five-year TCO	Management response
Controlled / -10%	\$21.1m	Reuse current finance/POS where supportable; strict configuration; staged license activation.
Baseline	\$23.4m	Deliver approved scope and Y4-5 run allowance.
Pressure / +15%	\$26.9m	Rephase optional capability; invoke contingency; require steering approval for scope expansion.

Commercial validation must include user/device/order-line metrics, non-production environments, storage/egress, premium connectors, partner support, annual indexation, implementation travel, refresh cycles, exit fees and internal backfill. Public list prices are directional only; Microsoft currently publishes Australian Commerce, SCM, IOM and Power BI rates, while many enterprise vendors quote by scope [3-6].

4. INTEGRATION REQUIREMENTS

4.1 Interfaces and technical dependencies

Contract each flow as a monitored business service

Flow	Pattern / latency	Controls and dependency
Product + price -> channels	Event/batch; effective-dated	Approval, catalogue completeness, tax/promotion rules, reject queue and channel reconciliation.
Stock -> availability -> channel	Event stream; normal <=5 min	Sequence/version, duplicates, location status, safety stock, lag alert and full resync.
Quote/ATP/reserve	Synchronous API	Timeout budget, reservation expiry, idempotency, capacity/cutoff and degraded-mode message.
Checkout/payment -> OMS	Synchronous command + event	Tokenised payment, fraud result, idempotent order, duplicate-charge prevention and audit.
OMS -> WMS/store	Event/command	Routing version, release/acknowledge, split/merge, cancellation boundary and exception owner.
Fulfilment -> carrier/customer	Event/API	Labels, manifest, tracking, proof, ETA updates and missed-scan alert.
Returns/refunds -> inventory/finance	Workflow + events	Disposition, serial/warranty, fraud, original tender, stock availability and balanced posting.
Customer/consent -> channels	Event/API	Identity, purpose, channel, timestamp, source, suppression and audit evidence.
Operational events -> data/BI	Streaming/batch	Schema registry, lineage, quality rules, late-arriving data and source-total reconciliation.

Technical standards

API-first, versioned OpenAPI contracts; OAuth 2.0/OIDC; managed identities; least privilege; secrets in a managed vault; rate limits and input validation.

Canonical identifiers for product, location, customer and order; UTC event time plus local business time; documented units, currency and status transitions.

End-to-end correlation, service-level objectives, synthetic journey tests, business reconciliation, dead-letter ownership and controlled replay.

Separate development/test/UAT/performance/training/production environments with masked or synthetic non-production data and repeatable deployment.

Release dependency The website may enter controlled release only after product/price, ATP/reservation, payment, OMS, fulfilment, status and return paths pass end-to-end tests—including partial failures and rollback. Interface completion alone is not acceptance.

DATA MIGRATION

4.2 Data migration plan

Domain ownership, rehearsals and reconciliation precede cutover

Domain	Migration treatment	Acceptance control
Product, hierarchy, attributes	Profile, deduplicate, enrich mandatory commerce/WMS fields; publish approved golden records.	Completeness by category; duplicate/invalid rate; sample visual/scan test.
Price, tax, promotion	Effective-date mapping and overlap/conflict resolution; migrate active/future rules.	Source-to-target totals and scenario price tests across web/POS.
Customer, consent, loyalty	Identity resolution with conservative merge; retain purpose, source, time and opt-out evidence.	Consent reconciliation, duplicate review and privacy-owner sign-off.
Inventory, locations, bins, serials	Clean location/bin master; freeze/count/snapshot plus controlled delta; preserve holds/reservations/status.	Physical count and valuation reconciliation within approved tolerance.
Open orders, returns, gift/loyalty balances	Migrate active items or operate a documented history lookup; map every status and financial liability.	100% control totals; sampled journey completion and balanced posting.
History and analytics	Load only justified history to operational systems; archive accessible records; create governed analytical history.	Record counts, aggregates, lineage, retention and access tests.

Migration waves

Wave	Activity	Exit gate
M1-3 Discover	Inventory sources, owners, volumes, quality, retention and dependencies; establish baselines.	Signed data catalogue and remediation plan.
M4-10 Prepare	Cleanse, map, build automated controls and complete two full rehearsals in production-like environments.	Defect trend, performance and totals within tolerance.
M11-18 Pilot	Snapshot + delta cutover for controlled web/WMS/store cohort; retain rollback and read-only legacy access.	Operational and reconciliation thresholds sustained.
M18-29 Scale	Repeatable store/category/location waves with entry/exit checklists and hypercare.	Each wave accepted before next cohort.
M30-36 Optimise	Archive/decommission where allowed; validate retention, deletion and support ownership.	Business owner and audit sign-off.

Cutover rule No migration proceeds on percentage-complete reporting alone. The data owner must approve named control totals, exceptions, business sampling, privacy/security checks, operational readiness and a timed rollback decision point.

SECURITY, PRIVACY AND DELIVERY

4.3 Security considerations and 5.2 implementation roadmap

Controls are built into stage gates and operating ownership

Area	Minimum requirement
Privacy	Privacy impact assessment; data minimisation; consent purpose and evidence; retention/deletion; customer access/correction; processor/subprocessor review; Australian Privacy Principles alignment.
Payments	Minimise PCI scope through tokenised/hosted payment patterns; no card data in logs or analytics; segmentation, vulnerability management and annual responsibility mapping.
Identity	MFA, SSO, least privilege, separation of duties, privileged access workflow, joiner/mover/leaver automation and quarterly access review.
Applications/APIs	Secure SDLC, threat modelling, SAST/DAST/dependency scans, penetration testing, OWASP API controls, rate limits and abuse/fraud monitoring.
Data	Classification, encryption, managed keys/secrets, masking/synthetic test data, immutable audit, DLP and monitored export.
Resilience	Service-specific RTO/RPO, multi-zone design where justified, backups, restore and failover tests, manual/degraded procedures and supplier incident obligations.
AI	Approved use case/data, human oversight, accuracy/bias/drift monitoring, prompt/output controls where relevant, model/version trace and safe fallback.

Implementation roadmap and complexity

Stage	Months	Technology outcome	Complexity / gate
Foundation	1-9	Landscape, process, data, API/event, identity, security, observability and BI baselines.	High Gate: architecture/data/control readiness
Build + controlled pilot	4-18	Commerce, OMS/ATP, WMS and representative stores; CRM foundation.	Very high Gate: golden orders + pilot thresholds
Scale	18-29	Store/location/category waves, customer service and operational dashboards.	High Gate: sustained accuracy, customer and labour results
Optimise	30-36	Demand-planning pilot, measured personalisation, support transfer and benefit closure.	Med-high Gate: data/model/value proof

Core delivery resources: programme architect; commerce, OMS, WMS, POS, CRM and data product owners; integration/data/security architects; business analysts; developers/configurators; UX/accessibility and content specialists; warehouse/store/process leads; testers/performance engineers; privacy/PCI specialists; change/training leads; local super users; service-management and vendor teams. Peak demand occurs in Months 6-18; internal backfill is part of TCO.

GOVERNANCE AND SOURCES

5.3 Decision gates, conclusion and references

Immediate decisions and evidence gates

Decision	Timing	Required evidence
Approve Microsoft-centred shortlist and specialist contingencies	Now	This assessment, architecture principles and procurement plan.
Confirm finance ERP coexist vs replace	M3	Fit-gap, support/security risk, integration economics and funded business case.
Confirm WMS/OMS choice	M6-10	Warehouse observation, volume/automation model, golden-order proof, references and binding TCO.
Approve controlled release	M11	Performance/accessibility/security/payment, data reconciliation, rollback and trained support.
Approve scale waves	M18 onward	Sustained accuracy, readiness, labour, customer and cost thresholds.
Approve AI scale	M30	Frozen back-test, planner usability, model risk, privacy and realised-value evidence.

Conclusion

TechHub should standardise on a Microsoft-centred retail and data platform while preserving disciplined specialist escape routes. The recommendation is justified by end-to-end process fit, lower integration burden, staged delivery inside the approved programme envelope and a credible operating model. Success depends less on the logo chosen than on verified data, explicit system ownership, monitored business flows, process-led pilots, security/privacy gates and contractual evidence. The ERP boundary and specialist WMS/forecasting contingencies must remain executive decisions—not hidden implementation drift.

Selected references

1. [TechHub Digital Transformation Roadmap - Revised \(internal\)](#) - programme scope, benefits, sequence and controls
2. [Microsoft Dynamics 365 Commerce documentation](#)
3. [Microsoft Dynamics 365 Commerce pricing \(Australia\)](#)
4. [Microsoft Dynamics 365 Supply Chain Management pricing \(Australia\)](#)
5. [Microsoft Power BI pricing \(Australia\)](#)
6. [Microsoft Dynamics 365 Inventory Visibility](#)
7. [Microsoft Dynamics 365 Commerce distributed order management](#)
8. [Microsoft Dynamics 365 Demand Planning](#)
9. [Salesforce Service Cloud pricing and editions](#)
10. [HubSpot Customer Platform](#)
11. [SAP S/4HANA Cloud](#)
12. [Oracle Fusion Cloud ERP](#)
13. [Adobe Commerce](#)
14. [Shopify Plus retail and POS](#)
15. [Oracle Retail Xstore and omnichannel](#)
16. [Manhattan Active Warehouse Management](#)
17. [Infor Warehouse Management System](#)
18. [Manhattan Active Order Management](#)
19. [Fluent Order Management](#)
20. [RELEX Forecasting and Replenishment](#)
21. [Blue Yonder Demand Planning](#)
22. [Tableau pricing](#)
23. [Qlik Cloud Analytics](#)
24. [NIST Cybersecurity Framework 2.0](#)
25. [OWASP API Security Top 10](#)
26. [PCI Security Standards Council](#)
27. [Office of the Australian Information Commissioner - Australian Privacy Principles](#)

Source note: vendor capabilities and public prices were reviewed from official pages. Accessed 7 September 2026. All vendor pricing, scope and scores remain subject to RFP and due diligence.